

- 281.** If the ratio of the fifth term from the beginning to the fifth term from the end in the expansion of $\left(\sqrt[4]{2} + \frac{1}{\sqrt[4]{3}}\right)^n$ is $\sqrt{6} : 1$, then the third term from the beginning is :

यदि $\left(\sqrt[4]{2} + \frac{1}{\sqrt[4]{3}}\right)^n$ के विस्तार में आरंभ से पाँचवें पद का अंत से पाँचवे पद से अनुपात $\sqrt{6} : 1$ है, तब आरंभ से तीसरा पद है :

- (A) $30\sqrt{3}$ (B) $60\sqrt{3}$ (C) $60\sqrt{2}$ (D) $30\sqrt{2}$

Ans. B

Sol.
$$\frac{{}^nC_4 2^{\frac{n-4}{4}} \cdot \left(3^{-\frac{1}{4}}\right)^4}{{}^nC_4 3^{-\left(\frac{n-4}{4}\right)} \cdot \left(2^{\frac{1}{4}}\right)^4} = \frac{\sqrt{6}}{1}$$

$$\Rightarrow n = 10$$

$$\text{So } T_3 = {}^{10}C_2 2^{\frac{1}{4} \cdot 8} \cdot 3^{-\frac{1}{4} \cdot 2} = \frac{45 \cdot 4}{\sqrt{3}} = 60\sqrt{3}$$

- 282.** If the coefficients of x^7 in $\left(ax^2 + \frac{1}{2bx}\right)^{11}$ and x^{-7} in $\left(ax - \frac{1}{3bx^2}\right)^{11}$ are equal, then :

यदि $\left(ax^2 + \frac{1}{2bx}\right)^{11}$ में x^7 तथा $\left(ax - \frac{1}{3bx^2}\right)^{11}$ में x^{-7} के गुणांक बराबर हैं, तो :

- (A) $64ab = 243$ (B) $243ab = 64$ (C) $729ab = 32$ (D) $32ab = 729$

Ans. C

Sol. Coefficient of x^7 in $\left(ax^2 + \frac{1}{2bx}\right)^{11}$

$$T_{r+1} = {}^{11}C_r (ax^2)^{11-r} \left(\frac{1}{2bx}\right)^r$$

$$= {}^{11}C_r (a)^{11-r} \left(\frac{1}{2b}\right)^r x^{22-3r}$$

$$22 - 3r = 7 \Rightarrow r = 5$$

$$\text{Coefficient of } x^{-7} \text{ in } \left(ax - \frac{1}{3bx^2}\right)^{11}$$

$$T_{r+1} = {}^{11}C_r (ax)^{11-r} \left(-\frac{1}{3bx^2} \right)^r$$

$$= {}^{11}C_r a^{11-r} \left(-\frac{1}{3b} \right)^r x^{11-3r}$$

$$\therefore 11-3r = -7 \Rightarrow r = 6$$

$$\Rightarrow {}^{11}C_5 (ab)^6 \left(\frac{1}{2b} \right)^5 = 32$$

$$\Rightarrow 729ab = 32$$

283. $25^{190} - 19^{190} - 8^{190} + 2^{190}$ is divisible by :

(A) 14 but not by 34 (B) both 14 and 34 (C) 34 but not by 14 (D) neither 14 nor 34

$$25^{190} - 19^{190} - 8^{190} + 2^{190}$$

(A) 14 से विभाज्य है परन्तु 34 से नहीं

(B) 14 तथा 34 दोनों से विभाज्य है

(C) 34 से विभाज्य है परन्तु 14 से नहीं

(D) न तो 14 से नहीं 34 से विभाज्य है

Ans. C

Sol. $(25^{190} - 19^{190}) - (8^{190} - 2^{190})$ is divisible by 6.

$(25^{190} - 8^{190}) - (19^{190} - 2^{190})$ is divisible by 17.

$25^{190} - 8^{190}$ is not divisible by 7

but $19^{190} - 2^{190}$ is divisible by 7

So, $25^{190} - 19^{190} - 8^{190} + 2^{190}$ is divisible by 34 but not 14

284. Let the number $(22)^{2022} + (2022)^{22}$ leave the remainder α when divided by 3 and β when divided by 7. Then $(\alpha^2 + \beta^2)$ is equal to :

माना $(22)^{2022} + (2022)^{22}$ को 3 से विभाजित करने पर शेषफल α है तथा 7 से विभाजित करने पर शेषफल β है। तो $(\alpha^2 + \beta^2)$ बराबर है :

(A) 13

(B) 5

(C) 20

(D) 10

Ans. B

Sol. For $\alpha : (21+1)^{2022} + \underbrace{(2022)^{22}}_{\text{divisible by 3}}$

$$= 3K_1 + 1$$

For $\beta : (21+1)^{2022} + (2023-1)^{22}$

$$= 7\lambda + 1 + 7\mu + 1$$

$$= 7K_2 + 2$$

So, $\alpha = 1, \beta = 2$

$$\alpha^2 + \beta^2 = 5$$

- 285.** If the 1011th term from the end in the binomial expansion of $\left(\frac{4x}{5} - \frac{5}{2x}\right)^{2022}$ is 1024 times 1011th term from the beginning, then $|x|$ is equal to :

यदि $\left(\frac{4x}{5} - \frac{5}{2x}\right)^{2022}$ के द्विपद प्रसार में अंत से 1011वाँ पद, आरंभ से 1011वें पद का 1024 गुना है, तो $|x|$ बराबर है :

- (A) 8 (B) 12 (C) 10 (D) 15

Ans. C

Sol. T_{1011} from beginning = T_{1010+1}

$$= {}^{2022}C_{1010} \left(\frac{4x}{5}\right)^{1012} \left(\frac{-5}{2x}\right)^{1010}$$

T_{1011} from end

$$= {}^{2022}C_{1010} \left(\frac{-5}{2x}\right)^{1012} \left(\frac{4x}{5}\right)^{1010}$$

$$\text{Given : } = {}^{2022}C_{1010} \left(\frac{-5}{2x}\right)^{1012} \left(\frac{4x}{5}\right)^{1010}$$

$$= 2^{10} \cdot {}^{2022}C_{1010} \left(\frac{-5}{2x}\right)^{1010} \left(\frac{4x}{5}\right)^{1012}$$

$$\left(\frac{-5}{2x}\right)^2 = 2^{10} \left(\frac{4x}{5}\right)^2$$

$$x^4 = \frac{5^4}{2^{16}}$$

$$|x| = \frac{5}{16}$$

- 286.** The sum, of the coefficients of the first 50 terms in the binomial expansion of $(1-x)^{100}$, is equal to :

$(1-x)^{100}$ के द्विपद प्रसार में प्रथम 50 पदों के गुणांकों का योग बराबर है :

- (A) ${}^{99}C_{49}$ (B) $-{}^{99}C_{49}$ (C) $-{}^{101}C_{50}$ (D) ${}^{101}C_{50}$

Ans. B

Sol. $(1-x)^{100} = C_0 - C_1x + C_2x^2 - C_3x^3 + \dots + C_{99}x^{99} + C_{100}x^{100}$

$$\Rightarrow C_0 - C_1 + C_2 - C_3 + \dots - C_{99} + C_{100} = 0$$

$$2(C_0 - C_1 + C_2 + \dots - C_9) + C_{50} = 0$$

$$C_0 - C_1 + C_2 + \dots + C_{99} = -\frac{1}{2} C_{100}$$

$$-\frac{1}{2} \frac{100!}{50!50!} = -\frac{1}{2} \times \frac{100 \times 99!}{50!50!} = -{}^{99}C_{49}$$

287. Let $(a + bx + cx^2)^{10} = \sum_{i=0}^{20} p_i x^i$, $a, b, c \in \mathbb{N}$. If $p_1 = 20$ and $p_2 = 210$, then $2(a+b+c)$ is equal to :

यदि $(a + bx + cx^2)^{10} = \sum_{i=0}^{20} p_i x^i$, $a, b, c \in \mathbb{N}$, $p_1 = 20$ तथा $p_2 = 210$ हो, तो $2(a+b+c)$ का मान बराबर होगा :

(A) 8

(B) 12

(C) 6

(D) 15

Ans. B

Sol. $(a + bx + cx^2)^{10} = \sum_{i=0}^{20} p_i x^i$,

Coefficient of $x^1 = 20$

$$20 = \frac{10!}{9!1!} \times a^9 \times b^1$$

$$a^9 \cdot b = 20$$

$$a = 1, b = 2$$

Coefficient of $x^2 = 210$

$$210 = \frac{10!}{9!1!} \times a^9 \times c^1 + \frac{10!}{8!2!} \times a^8 b^2$$

$$210 = 10 \cdot c + 45 \times 4$$

$$10c = 30$$

$$c = 3$$

$$2(a + b + c) = 12$$

288. If $\sum_{r=0}^{10} \left(\frac{10^{r+1} - 1}{10^r} \right) \cdot {}^{11}C_{r+1} = \frac{\alpha^{11} - 11^{11}}{10^{10}}$, then α is equal to :

यदि $\sum_{r=0}^{10} \left(\frac{10^{r+1} - 1}{10^r} \right) \cdot {}^{11}C_{r+1} = \frac{\alpha^{11} - 11^{11}}{10^{10}}$ है, तो α बराबर है :

(A) 15

(B) 20

(C) 24

(D) 11

Ans. B

Sol. $\sum_{r=0}^{10} \left(\frac{10^{r+1} - 1}{10^r} \right) \cdot {}^{11}C_{r+1}$

$$\begin{aligned}
 &= (10-1) {}^{11}C_1 + \left(\frac{10^2-1}{10}\right) {}^{11}C_2 + \dots + \left(\frac{10^{11}-1}{10^{10}}\right) {}^{11}C_{11} \\
 &= 10({}^{11}C_1 + {}^{11}C_2 + \dots + {}^{11}C_{11}) - \left({}^{11}C_1 + \frac{{}^{11}C_2}{10} + \dots + \frac{{}^{11}C_{11}}{10^{10}}\right) \\
 &= 10(2^{11}-1) - 10\left(\frac{{}^{11}C_1}{10} + \frac{{}^{11}C_2}{10^2} + \dots + \frac{{}^{11}C_{11}}{10^{11}}\right) \\
 &= 10(2^{11}-1) - 10\left(\left(1 + \frac{1}{10}\right)^{11} - 1\right) \\
 &= 10(2^{11}-1) - 10\left(\frac{11^{11}-10^{11}}{10^{11}}\right) \\
 &= 10(2^{11}-1) - \left(\frac{11^{11}-10^{11}}{10^{10}}\right) \\
 &= \frac{10^{11}(2^{11}-1) - 11^{11} + 10^{11}}{10^{10}} \\
 &= \frac{2^{11}(10^{11}) - 11^{11}}{10^{10}} = \frac{20^{11} - 11^{11}}{10^{10}} \\
 &\Rightarrow \alpha = 20
 \end{aligned}$$

- 289.** If the sum of the coefficients of all even powers of x in the product $(1+x+x^2+\dots+x^{2n})(1-x+x^2-x^3+\dots+x^{2n})$ is 61, then n is equal to _____.

यदि गुणनफल $(1+x+x^2+\dots+x^{2n})(1-x+x^2-x^3+\dots+x^{2n})$ में x के सभी सम-घातों वाले गुणांकों का योगफल 61 है, तो n बराबर है _____।

Ans. 30

Sol. Let $(1+x+x^2+\dots+x^{2n})(1-x+x^2-x^3+\dots+x^{2n})$

$$= a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_{4n}x^{4n} \quad \dots\dots\dots(1)$$

Put $x=1$ in equation (1)

$$2n+1 = a_0 + a_1 + a_2 + a_3 + \dots + a_{4n} \quad \dots\dots\dots(2)$$

Put $x=-1$ in equation (1)

$$2n+1 = a_0 - a_1 + a_2 - a_3 + \dots + a_{4n} \quad \dots\dots\dots(3)$$

Adding (2) and (3)

$$2(a_0 + a_2 + a_4 + \dots + a_{4n}) = 4n + 2$$

$$\Rightarrow 2 \times 61 = 2(2n+1) \quad \Rightarrow 2n+1 = 61 \quad \Rightarrow n = 30$$

290. Let $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$. Then $\frac{a_7}{a_{13}}$ is equal to _____.

माना $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$, तो $\frac{a_7}{a_{13}}$ बराबर है _____।

Ans. 8

Sol. $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$

General term of $(2x^2 + 3x + 4)^{10}$

$$= \frac{10!}{p_1! p_2! p_3!} (2x^2)^{p_1} (3x)^{p_2} (4)^{p_3}$$

Where $p_1 + p_2 + p_3 = 10$

and $0 \leq p_i \leq 10; p_i \in W, i = 1, 2, 3$

a_7 = coefficient of x^7

a_{13} = coefficient of x^{13} .

$$= \frac{10!}{p_1! p_2! p_3!} 2^{p_1} 3^{p_2} 4^{p_3} x^{2p_1 + p_2}$$

For a_7 : $2p_1 + p_2 = 7$

p_1	p_2	p_3	Coefficient of x^7
3	1	6	$\frac{10!}{3!1!6!} 2^3 \cdot 3^1 \cdot 4^6 = \frac{10!}{3!1!6!} 2^{15} \cdot 3^1$
2	3	5	$\frac{10!}{2!3!5!} 2^2 \cdot 3^3 \cdot 4^5 = \frac{10!}{2!3!5!} 2^{12} \cdot 3^3$
1	5	4	$\frac{10!}{1!5!4!} 2^1 \cdot 3^5 \cdot 4^4 = \frac{10!}{1!5!4!} 2^9 \cdot 3^5$
0	7	3	$\frac{10!}{7!3!} 1 \cdot 3^7 \cdot 4^3 = \frac{10!}{7!3!} 2^6 \cdot 3^7$

$$a_7 = 2^3 \left(\frac{10!}{3!1!6!} 2^{12} \cdot 3^1 + \frac{10!}{2!3!5!} 2^9 \cdot 3^3 + \frac{10!}{1!5!4!} 2^6 \cdot 3^5 + \frac{10!}{7!3!} 2^3 \cdot 3^7 \right)$$

For a_{13} : $2p_1 + p_2 = 13$

p_1	p_2	p_3	Coefficient of x^{13}
6	1	3	$\frac{10!}{6!.1!.3!} 2^6 \cdot 3^1 \cdot 4^3 = \frac{10!}{6!.1!.3!} 2^{12} \cdot 3^1$
5	3	2	$\frac{10!}{5!.3!.2!} 2^5 \cdot 3^3 \cdot 4^2 = \frac{10!}{5!.3!.2!} 2^9 \cdot 3^3$
4	5	1	$\frac{10!}{4!.5!.1!} 2^4 \cdot 3^5 \cdot 4^1 = \frac{10!}{4!.5!.1!} 2^6 \cdot 3^5$
3	7	0	$\frac{10!}{3!.7!} 2^3 \cdot 3^7 = \frac{10!}{3!.7!} 2^3 \cdot 3^7$

$$a^{13} = \left(\frac{10!}{6!.1!.3!} 2^{12} \cdot 3^1 + \frac{10!}{5!.3!.2!} 2^9 \cdot 3^3 + \frac{10!}{4!.5!.1!} 2^6 \cdot 3^5 + \frac{10!}{3!.7!} 2^3 \cdot 3^7 \right)$$

$$\frac{a_7}{a_{13}} = 2^3 = 8$$

- 291.** The natural number m , for which the coefficient of x in the binomial expansion of $\left(x^m + \frac{1}{x^2}\right)^{22}$ is 1540, is ____.

प्राकृत संख्या m , जिसके लिए $\left(x^m + \frac{1}{x^2}\right)^{22}$ के द्विपद प्रसार में x का गुणांक 1540 है, है is ____ ।

Ans. 13

Sol. General term $T_{r+1} = {}^{22}C_r (x^m)^{22-r} \cdot \left(\frac{1}{x^2}\right)^r$

$$= {}^{22}C_r x^{22m-mr-2r}$$

$$22m - mr - 2r = 1 \dots\dots\dots (1)$$

$$\text{and } {}^{22}C_r = 1540$$

$${}^{22}C_3 = \frac{22 \times 21 \times 20}{3 \times 2 \times 1} = 1540$$

So $r = 3$ or 19 .

$$\text{If } r = 3, 22m - 3m - 6 = 1$$

$$\Rightarrow m = \frac{7}{19} \text{ (not an integer)}$$

$$\text{If } r = 19, 22m - 19m - 38 = 1$$

$$\Rightarrow m = 13$$

- 292.** If the remainder when x is divided by 4 is 3, then the remainder when $(2020 + x)^{2022}$ is divided by 8 is ____ .

यदि x को 4 से विभाजित करने पर शेषफल 3 है, तो $(2020 + x)^{2022}$ को 8 से विभाजित करने पर शेषफल है _____ ।

Ans. 1

Sol. $x = 4k + 3$

$$(2020 + x)^{2022} = (2023 + 4k)^{2022}$$

$$(2024 + (4k - 1))^{2022}$$

$$= {}^{2022}C_0 (2024)^{2022} + \dots + {}^{2022}C_{2021} (2024)(4k-1)^{2021} + {}^{2022}C_{2022} (4k-1)^{2022}$$

$$= 8I + {}^{2022}C_{2022} (4k-1)^{2022}$$

$$= 8I + (4k-1)^{2022}$$

$$= 8I + {}^{2022}C_0 (4k)^{2022} + \dots + {}^{2022}C_{2022} (-1)^{2022}$$

$$= 8I + 8I_1 + 1$$

$$\text{Remainder} = 1$$

293. If $(2021)^{3762}$ is divided by 17, then the remainder is _____.

यदि $(2021)^{3762}$ को 17 से विभाजित करने पर शेषफल _____ ।

Ans. 4

Sol. $(2021)^{3768} = (2023 - z)^{3762}$

$$= 17k + 2^{3762} \quad \{k \in I\}$$

$$= 17k + 2^2 (2^4)^{940}$$

$$= 17k + 4 (17-1)^{940}$$

$$= 17k + 4 (17\lambda + 1) \quad \{\lambda \in I\}$$

$$= 17k + 4 (17\lambda) + 4$$

$$\text{Remainder} = 4$$

294. Let n be a positive integer. Let $A = \sum_{k=0}^n (-1)^k n_{C_k} \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k \right]$. If $63A = 1 - \frac{1}{2^{30}}$,

then n is equal to :

माना n एक धनात्मक पूर्णांक है तथा $A = \sum_{k=0}^n (-1)^k n_{C_k} \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k \right]$ है। यदि

$63A = 1 - \frac{1}{2^{30}}$ है, तो n बराबर है _____ ।

Ans. 6

Sol. $A = \sum_{k=0}^n (-1)^k n_{C_k} \left[\left(\frac{1}{2} \right)^k + \left(\frac{3}{4} \right)^k + \left(\frac{7}{8} \right)^k + \left(\frac{15}{16} \right)^k + \left(\frac{31}{32} \right)^k \right]$

$$A = \left(1 - \frac{1}{2} \right)^n + \left(1 - \frac{3}{4} \right)^n + \left(1 - \frac{7}{8} \right)^n + \left(1 - \frac{15}{16} \right)^n + \left(1 - \frac{31}{32} \right)^n$$

$$A = \frac{1}{2^n} + \frac{1}{4^n} + \frac{1}{8^n} + \frac{1}{16^n} + \frac{1}{32^n}$$

$$A = \frac{1}{2^n} + \frac{1}{2^{2n}} + \frac{1}{2^{3n}} + \frac{1}{2^{4n}} + \frac{1}{2^{5n}}$$

$$A = \frac{\frac{1}{2^n} \left(\left(\frac{1}{2^n} \right)^5 - 1 \right)}{\left(\frac{1}{2^n} - 1 \right)}$$

$$A = \frac{1 - 2^{5n}}{2^{5n}(1 - 2^n)} = \frac{2^{5n} - 1}{2^{5n}(2^n - 1)}$$

$$63A = \left(\frac{63}{2^n - 1} \right) \left(1 - \frac{1}{2^{5n}} \right) = 1 - \frac{1}{2^{30}}$$

295. If the constant term in the binomial expansion of $\left(\frac{x^{\frac{5}{2}}}{2} - \frac{4}{x^\ell} \right)^9$ is -84 and the coefficient of $x^{-3\ell}$ is $2^\alpha \beta$, where

$\beta < 0$ is an odd number, then $|\alpha\ell - \beta|$ is equal to _____.

यदि $\left(\frac{x^{\frac{5}{2}}}{2} - \frac{4}{x^\ell} \right)^9$ के द्विपद प्रसार में अचर पद -84 है तथा $x^{-3\ell}$ का गुणांक $2^\alpha \beta$ है, जहाँ $\beta < 0$ एक विषम संख्या है, तो $|\alpha\ell - \beta|$

बराबर है।

Ans. 98

Sol. In, $\left(\frac{x^{\frac{5}{2}}}{2} - \frac{4}{x^\ell} \right)^9$

$$T_{r+1} = {}^9C_r \frac{(x^{5/2})^{9-r}}{2^{9-r}} \left(\frac{-4}{x^l}\right)^r$$

$$= (-1)^r \frac{{}^9C_r}{2^{9-r}} 4^r x^{\frac{45}{2} - \frac{5r}{2} - lr}$$

$$= 45 - 5r - 2lr = 0$$

$$r = \frac{45}{5+2l} \quad \text{_____ (1)}$$

Now, according to the question, $(-1)^r \frac{{}^9C_r}{2^{9-r}} 4^r = -84$

$$= (-1)^r {}^9C_r 2^{3r-9} = 21 \times 4$$

Only natural value of r possible if $3r - 9 = 0$

$$r = 3 \text{ and } {}^9C_3 = 84$$

$$\therefore l = 5 \text{ from equation (1)}$$

Now, coefficient of $x^{-3l} = x^{\frac{45}{2} - \frac{5r}{2} - lr}$ at $l = 5$, gives

$$r = 5$$

$$\therefore {}^9C_5 (-1) \frac{4^5}{2^4} = 2^\alpha \times \beta$$

$$= -63 \times 2^7$$

$$\Rightarrow \alpha = 7, \beta = -63$$

$$\therefore \text{value of } |\alpha l - \beta| = 98$$

- 296.** Let the sum of the coefficients of the first three terms in the expansion of $\left(x - \frac{3}{x^2}\right)^n$, $x \neq 0$, $n \in \mathbb{N}$, be 376. Then the coefficients of x^4 is _____.

माना $\left(x - \frac{3}{x^2}\right)^n$, $x \neq 0$, $n \in \mathbb{N}$ के प्रसार में प्रथम तीन पदों के गुणांकों का योग 376 है। तो x^4 का गुणांक _____ है।

Ans. 405

Sol. ${}^nC_0 - {}^nC_1 \cdot 3 + {}^nC_2 \cdot 3^2 = 376$

$$\Rightarrow n = 10$$

Now $T_{r+1} = {}^{10}C_r \cdot \left(\frac{-3}{x^2}\right)^r \cdot (x)^{10-r}$

$$= {}^{10}C_r \cdot (-3)^r \cdot x^{10-3r}$$

For $x^4 \Rightarrow 10 - 3r = 4 \Rightarrow r = 2$

Coff. of $x^4 = {}^{10}C_2 \cdot (-3)^2 = 405$

297. The constant term in the expansion of $\left(2x + \frac{1}{x^7} + 3x^2\right)^5$ is _____.

$\left(2x + \frac{1}{x^7} + 3x^2\right)^5$ के प्रसार में अचर पद है।

Ans. 1080

Sol. General term is $\sum \frac{5!(2x)^{n_1} (x^{-7})^{n_2} (3x^2)^{n_3}}{n_1! n_2! n_3!}$

For constant term,

$$n_1 + 2n_3 = 7n_2$$

$$\& n_1 + n_2 + n_3 = 5$$

Only possibility $n_1 = 1, n_2 = 1, n_3 = 3$

$\Rightarrow$ constant term = 1080

298. The remainder when $(2023)^{2023}$ is divided by 35 is _____.

$(2023)^{2023}$ को 35 से विभाजित करने पर शेषफल है।

Ans. 7

Sol. $-72023 = -7 \times 7^{2022}$

$$= \frac{-7^{2022}}{5}$$

$$= \frac{-[50-1]^{1011}}{5} = \frac{[5\lambda-1]}{5}$$

remainder = + 1

when divided by 5 now remainder is 7 when divided 35

299. Let $\sum_{n=0}^{\infty} \frac{n^3 ((2n)!) + (2n-1)(n!)}{(n!)((2n!))} = ae + \frac{b}{e} + c$, where $a, b, c \in \mathbb{Z}$ and $e = \sum_{n=0}^{\infty} \frac{1}{n!}$. Then $a^2 - b + c$ is equal to

माना $\sum_{n=0}^{\infty} \frac{n^3 ((2n)!) + (2n-1)(n!)}{(n!)((2n!))} = ae + \frac{b}{e} + c$, है जहाँ $a, b, c \in \mathbb{Z}$ तथा $e = \sum_{n=0}^{\infty} \frac{1}{n!}$ है। तो $a^2 - b + c$ बराबर है।

Ans. 26

Sol.
$$\sum_{n=0}^{\infty} \frac{n^3 (2n)!}{n! (2n)!} + \sum_{n=0}^{\infty} \frac{(2n-1)n!}{n! (2n)!}$$

$$\sum_{n=0}^{\infty} \frac{(2n-1)n!}{n! (2n)!}$$

$$\sum_{n=0}^{\infty} \frac{(2n) \times n!}{n! (2n)!} - \frac{n!}{n! (2n)!}$$

$$\sum_{n=0}^{\infty} \frac{1}{(2n-1)!} - \sum_{n=0}^{\infty} \frac{1}{(2n)!}$$

$$\left. \frac{e^x - e^{-x}}{2} \right|_{x=1} - \left. \frac{e^x + e^{-x}}{2} \right|_{x=1}$$

$$= \sum_{n=0}^{\infty} \frac{n^3 (2n)!}{n! (2n)!} = \sum_{n=0}^{\infty} \frac{n^3}{n!}$$

$$n^3 = an(n-1)(n-2) + bn(n-1) + cn + d$$

$$n = 1 \Rightarrow c = 1$$

$$n = 2 \Rightarrow 8 = 2b + 2 \Rightarrow b = 3$$

$$= \sum_{n=0}^{\infty} \frac{n(n-1)(n-2) + 3n(n-1) + n}{n!}$$

$$\sum_{n=3}^{\infty} \frac{1}{(n-3)!} + 3 \sum_{n=2}^{\infty} \frac{1}{(n-2)!} + \sum_{n=1}^{\infty} \frac{1}{(n-1)!}$$

$$= \sum_{n=0}^{\infty} \frac{n^3 (2n)!}{n! (2n)!} + \sum_{n=0}^{\infty} \frac{(2n-1)n!}{n! (2n)!}$$

$$= 5e + \frac{e}{2} - \frac{1}{e} - \frac{e}{2} + \frac{1}{e}$$

$$= 5e - \frac{1}{e}$$

$$a = 5, b = -1, c = 0$$

$$\text{Now } a^2 - b + c = 26$$

- 300.** 50th root of a number x is 12 and 50th root of another number y is 18. Then the remainder obtained on dividing $(x + y)$ by 25 is _____.

एक संख्या x का 50वाँ मूल 12 है एवं दूसरी संख्या y का 50वाँ मूल 18 है। तब $(x + y)$ को 25 से विभाजित करने पर प्राप्त शेषफल है।

Ans. 23**Sol.** $(x)^{\frac{1}{50}} = 12$ and $(y)^{\frac{1}{50}} = 18$

$$\Rightarrow x = 12^{50} \Rightarrow y = 18^{50}$$

$$x + y = 12^{50} + 18^{50}$$

$$(2^2 \times 3^1)^{50} + (3^2 \times 2)^{50}$$

$$= 6^{50} (2^{50} + 3^{50})$$

$$= (5 + 1)^{50} (4^{25} + 9^{25})$$

$$= (25\lambda + 1) [(5 - 1)^{25} + (10 - 1)^{25}]$$

$$= (25\lambda + 1) (25k - 1 + 25\mu - 1)$$

$$= (25\lambda + 1) (25t + 23)$$

$$= 25\lambda + 23$$

so remiander is 23, when $x + y$ is divided by 25.